

Multiple Choice (10 marks)

1. To verify a digital signature we need the
 - (a) Sender's Private key
 - (b) Sender's Public key
 - (c) Receiver's Private key
 - (d) Receiver's Public key
2. In a _____ attack, the extra data that holds some specific instructions in the memory for actions is projected by a cyber-criminal or penetration tester to crack the system.
 - (a) Phishing
 - (b) MiTM
 - (c) Buffer-overflow
 - (d) Clickjacking
3. The _____ is categorized as an unknown segment of the Deep Web which has been purposely kept hidden & is inaccessible using standard web browsers.
 - (a) Haunted web
 - (b) World Wide Web
 - (c) Dark web
 - (d) Surface web
4. Which of the following does authorization aim to accomplish?
 - (a) Restrict what operations/data the user can access
 - (b) Determine if the user is an attacker
 - (c) Flag the user if he/she misbehaves
 - (d) Determine who the user is
5. _____ is the anticipation of unauthorized access or break to computers or data by means of wireless networks.
 - (a) Wireless access
 - (b) Wireless security
 - (c) Wired Security
 - (d) Wired device apps

True / False (10 marks)

Answer True or False

6. True or False: Memory leakage is a coding mistake that is primarily caused by inadequate processing power in applications.
7. True or False: The Metasploit framework simplifies the exploitation of vulnerabilities through a user-friendly, point-and-click interface.
8. True or False: The one-time pad key can be computed using the formula $k = m \text{ xor } c$.
9. True or False: Nmap classifies port states as active, inactive, or standby when assessing network connections.
10. True or False: DaCryptic is widely recognized as a remote Trojan similar to Troya in functionality.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the four-step authentication process that occurs between a wireless user and an access point, commonly known as the 4-way handshake, and analyze its significance in cybersecurity.
12. Discuss the principles of mutation-based fuzzing in cybersecurity, focusing on how it generates new inputs by modifying existing ones and its implications for software testing and vulnerability discovery.

End of Paper